Machine Learning Assignment – Units III to V

Roll number(1-10)

1. Explain the maximum-margin hyperplane and support vectors, and how soft margin (parameter C) handles overlapping data.
2. Explain the kernel trick with the polynomial and RBF kernels, using an example of a 1-D dataset made separable.

Roll number(11-20)

1. For TP = 45, FP = 10, FN = 5, TN = 40, compute accuracy, precision, recall and F1, and explain ROC–AUC.
2. Compare partitioned, hierarchical and density-based clustering with one example algorithm each.

Roll number(21-30)

1. Apply K-Means (K = 2) for two iterations to (1,1), (1,2), (3,4), (5,7), (3,5), (4,5), starting from centroids (1,1) and (5,7).
2. Why does K-Means fail on categorical data? Explain the K-Modes algorithm with a small example.

Roll number(31-40)

1. Explain the architecture and training steps of a Self-Organising Map, with applications.
2. Explain the E-step and M-step of EM for a Gaussian Mixture Model, and compare it with K-Means.

Roll number(41-50)

1. Explain the steps of PCA and find the eigenvalues of the covariance matrix [[4, 2], [2, 3]].
2. Explain Kernel PCA and t-SNE, and state how t-SNE differs from PCA.

Roll number(51-60)

1. Derive the bias-variance decomposition of error and explain how ensembles manage the trade-off.
2. Explain bagging and how Random Forest reduces variance, with OOB error and feature importance.

Roll number(61-LE-6)



1. Explain gradient boosting and the key features that make XGBoost fast and accurate.
2. Compare bagging and boosting, and explain how cross-validation and tuning reduce ensemble error.